ResumeLens

Does a résumé satisfy the formal pattern of a profile?

Daniel Barragan · Jacobo Rodriguez
Nicolas Mejia
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The problem

- One skill, many spellings: **JS**, **React.js**, **NodeJS**.
- Scores hide *why* a candidate matches.
- ResumeLens checks a pattern. It **never ranks** people.

Five steps, five formal models

1 Extract
Regular expressions
`text → tokens`

2 Normalize
Finite-state transducers
`tokens → canonical names`

3 Sort
Canonical order per profile
`names → sequence`

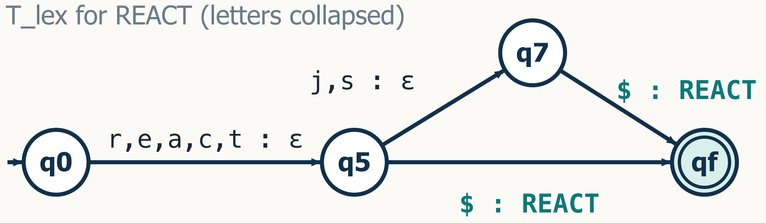
4 Recognize
DFA and ε-NFA
`sequence → verdict`

5 Report
Domain-specific language
`verdict → HTML / Markdown`

Normalize: transducers (FST)



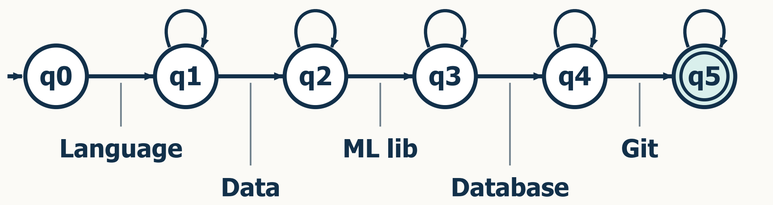
T_lex for REACT (letters collapsed)



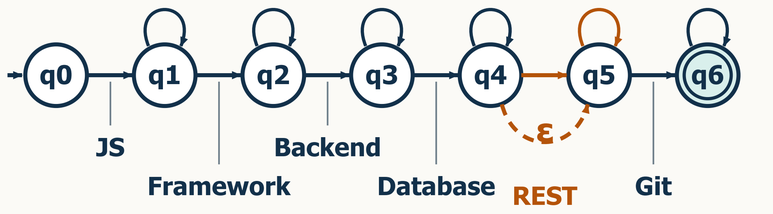
A Finite-State Transducer (FST) rewrites text. At the end marker **$** it prints the canonical name.

Recognize: finite automata

Machine Learning Engineer — **DFA**



Full Stack Developer — **ε-NFA**



Loops: several skills per category. Dashed **ε**: optional category skipped.

One résumé, four verdicts

`JS · React.js · NodeJS · Postgres · Git`

`JAVASCRIPT` `REACT` `NODE_JS` `POSTGRESQL` `GIT`

Full Stack	ε-NFA	ACCEPTED
ML Engineer	DFA · 3 missing	REJECTED
DevOps	ε-NFA · 5 missing	REJECTED
Data Engineer	DFA · 4 missing	REJECTED

`Full Stack path: q0→q1→q2→q3→q4 ε→q5→q6`

Validated with a DSL (textX)

```
candidate "Wednesday Addams" {
  classification {
    profile full_stack: ACCEPTED
      sequence [JAVASCRIPT, REACT, …]
  }
}
```

A Domain-Specific Language (DSL) grammar validates each candidate. Invalid text is rejected.

Take-aways

- Each step is a testable formal model.
- One résumé can match several profiles.

References. Hopcroft, Motwani & Ullman (2006). *Introduction to Automata Theory, Languages, and Computation*. Pearson. · Mohri (1997). Finite-state transducers in language and speech processing. *Computational Linguistics*, 23(2). · Romero (2021). Pyformlang. *SIGCSE '21*. · Dejanović et al. (2017). TextX. *Knowledge-Based Systems*, 115. · All diagrams were created by the authors.

Try it live
resumelens-fd9gqbfna3nbsnstxjte3i.streamlit.app
Code: github.com/Dabg5/ResumeLens